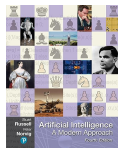


Lesson 08.4:

Instructor: Dr. GP Saggese, gsaggese@umd.edu

References:

- AIMA (Artificial Intelligence: a Modern Approach)
- Pearl et al., The Book of Why, 2017
- Facure, Causal Inference in Python, 2023



THIRD INTERNATIONAL EDITION / COURSEWARE PRESENTATION
JUDEA PEARL
WINNER OF THE TURING AWARD
AND DANA MACKENZIE
THE
BOOK OF
WHY
THE NEW SCIENCE
OF CAUSE AND EFFECT

- *Introduction to Causal Inference*
- Randomized Experiments and Stats Review
- Graphical Causal Models
- The Unreasonable Effectiveness of Linear Regression
- Propensity score
- Effect heterogeneity
- Metalerners
- Difference-in-differences
- Synthetic control
- Geo and switchback experiments
- Non-compliance and instruments
- Next steps

Sales example

- There are stores that put product on sales for Christmas

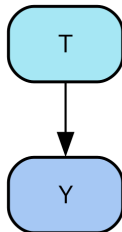
	store	weeks_to_xmas	avg_week_sales	is_on_sale	weekly_amount_sold
0	1	3	12.98	1.0	219.60
1	1	2	12.98	1.0	184.70
2	1	1	12.98	1.0	145.75
3	1	0	12.98	0.0	102.45
4	2	3	19.92	0.0	103.22
5	2	2	19.92	0.0	53.73

where

- `store`: id of the store
 - `weeks_to_xmas`: how many weeks before Christmas the measurement was done
 - `avg_week_sales`: how much they sold in average
 - `is_on_sale`: if store puts product on sale
 - `weekly_amount_on_sale`: how much that store has sold in that week
- Causal inference question: *“what is the impact of T on Y ?”*
 - E.g., what is the impact of `is_on_sale` on `weekly_amount_sold`?
 - E.g., when is the best moment to have sales (i.e., `weeks_to_xmas`)? ^{3 / 66}

Definition

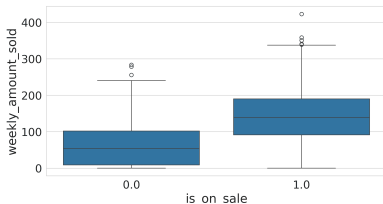
- **Treatment T**
 - $T_i = 1$ if unit i has received the treatment
 - Sometimes T is denoted with D
 - E.g., `is_on_sale` at `weeks_to_xmas`, `is_on_sale`, `received_discount`, `saw_ad`
- **Outcome Y**
 - The variable one wants to influence, Y_i
 - E.g., `weekly_amount_sold`
- **Unit of analysis**
 - The entity for which outcomes are measured



Study	Unit of analysis
Drug experiment	Patient
Marketing experiment	Customer
Sales promotion	Store or product
Education policy	Student

The Fundamental Problem of Causal Inference

- **The Fundamental Problem of Causal Inference**
 - “You can never observe the same unit with and without treatment”
- **Example**
 - Plot outcome
`weekly_amount_sold`
conditioned on treatment
`is_on_sale`
 - Stores with price drops sell more (from 50 to 150)
 - Matches intuition: “people buy more when prices are low due to sales”
- **Is it true?**
 - Maybe only larger businesses afford aggressive price drops?
 - Maybe customers buy close to Christmas anyway?
- Observing both counterfactual situations would reveal the effect of price drops (this is **impossible**)



Causal Model

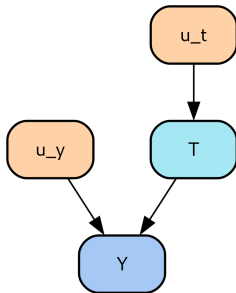
- A **causal model** is a series of assignments $\leftarrow$
 - Causality is non-reversible: you can't rearrange the equation

- **Example**

$$\begin{cases} T \leftarrow f_t(u_t) \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

where:

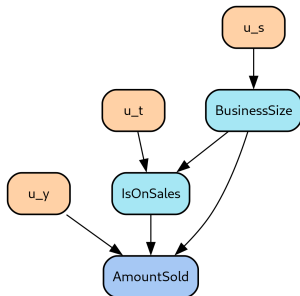
- u_t, u_y are *exogenous* variables (outside the model)
 - Account for everything not covered by model variables
- Treatment T and u_y cause outcome Y
- T and Y are *endogenous* variables (inside the model)



Causal Model: Sales Example

- Causal model for sales example:

$$\begin{cases} BusinessSize \leftarrow f_s(u_s) \\ IsOnSales \leftarrow f_t(BusinessSize, u_t) \\ AmountSold \leftarrow f_y(IsOnSales, BusinessSize, u_y) \end{cases}$$



- One can make an assumption of linear model for $f_y(\cdot)$ and say

$$AmountSold = \alpha + \beta_1 IsOnSales + \beta_2 BusinessSize + \varepsilon$$

Applying a Treatment to Sales Example

- Assume the **causal model** is:

$$\begin{cases} T \leftarrow f_t(u_t) \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

- Ask “what would happen to the outcome Y if I were to set the treatment to t_0 ?”

$$\begin{cases} T \leftarrow t_0 \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

- The intervention is represented by the $do()$ operator, e.g.,
 $do(T = t_0)$
- Note that **conditioning** and **intervention** are different:

$$\mathbb{E}[\text{AmountSold} | \text{isOnSales} = 1] \neq \mathbb{E}[\text{AmountSold} | do(\text{IsOnSales} = 1)]$$

- The **first** is what happens to businesses that make a price cut
 - Some business (e.g., big businesses) are “self-selecting” to cut prices
- The **second** is what happens if you force every business to engage in sales

Individual Treatment Effect

- Express the impact of the treatment T on the outcome Y for an individual unit i

$$\tau_i = Y_i|do(T = t_1) - Y_i|do(T = t_0)$$

- The effect τ_i of going from treatment t_0 to t_1 for unit i is the difference in the outcome of that unit under t_1 compared to t_0
- E.g.,
 $AmountSold_i|do(IsOnSales = 1) - AmountSold_i|do(IsOnSales = 0)$
- You can only observe one term due to the fundamental problem of causal inference
 - Represent it in theory, but can't recover it from data

Potential Outcomes

- Aka “counterfactuals”
- The $do()$ operator represents “unit i outcome is Y if treatment is set to t ”

$$Y_{t,i} \triangleq Y_i | do(T_i = t)$$

- The first is “Rubin’s notation”, the second is “Pearl’s notation”
- For binary treatment use Y_{0i} (without treatment) vs Y_{1i} (with treatment)
 - One is “factual” (i.e., observable)
 - The other is “counterfactual” (i.e., theoretical, not observable)
- There is a slight difference
 - $Y_t = Y | do(T = t)$ is “what a business would sell if it cut prices”
 - $Y | T = t$ is “what a business that cut the prices sold”

Consistency of the Treatment

- There is a single version of the treatment
 - I.e., $T = T_i$ means only one thing
- E.g.,
 - A drug can come in multiple dosages: only one is used
 - Giving a discount coupon is treated as the value of the coupon is the same, it's given only once, in a way that is always equivalent
 - In “help from a financial planner”, is “help” a one-time consultation or regular advice?

No Interference

- Aka “Stable Unit of Treatment Value (SUTVA)”
- The effect of one unit is not influenced by the treatment of other units
 - E.g., there are no spillovers or network effects
- E.g., vaccinating one person makes other close people less likely to catch the illness even if they don't get the treatment

Average Treatment Effects

- Assume you can see both potential outcomes Y_{1i} and Y_{0i} at the same time (which is impossible)
- **Average Treatment Effect (ATE)**
 - The impact that the treatment T would have on the average:

$$ATE = \mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[Y|do(T = 1)] - \mathbb{E}[Y|do(T = 0)]$$

- E.g., average impact of price cuts on amount sold:

$$ATE = \mathbb{E}[AmountSold_{1i} - AmountSold_{0i}]$$

Conditional Average Treatment Effects (CATE)

- The impact of the treatment on the units in a group defined by $X = x$

$$CATE = \mathbb{E}[(Y_{1i} - Y_{0i})|X = x]$$

- E.g., the impact of having sales during the week of Christmas:

$$CATE = \mathbb{E}[AmountSold_{1i} - AmountSold_{0i} | weeksToXmas = 0]$$

- Like ATE but on a subgroup with a certain characteristics

Average Treatment effect on the Treated (ATT)

- The impact of the treatment on the units that got the treatment:

$$ATT = \mathbb{E}[(Y_{1i} - Y_{0i}) | T = 1]$$

- E.g., how much sales increase for the business engaged in price cuts:

$$ATT = \mathbb{E}[(AmountSold_{1i} - AmountSold_{0i}) | IsOnSales = 1]$$

- We can define these quantities ATE , $CATE$, ATT , but not **measure them** directly
 - Because we can't observe both potential outcomes Y_{1i} and Y_{0i} at the same time

Bias

- **Bias** is what makes association different from causation
- The claim *“cutting prices increases the amount sold by a business”* can be questioned with:
 - *“Those businesses that cut prices could probably have sold more anyway, even with no price cuts”*
- More formally the argument refers to potential outcomes
 - We are trying to estimate $\mathbb{E}[Y_{it}] \triangleq \mathbb{E}[Y_i | do(T = t)]$ using $\mathbb{E}[Y_i | T = t]$ but this is a “biased estimator” (i.e., an incorrect one)

The Bias Equation

- **Causal inference** is interested in:

$$\mathbb{E}[Y_1 - Y_0] = \mathbb{E}[Y|do(T = 1)] - \mathbb{E}[Y|do(T = 0)]$$

- **What is measured** is:

$$\mathbb{E}[Y|T = 1] - \mathbb{E}[Y|T = 0]$$

- Let's express one in terms of the other
 - Replacing the observed outcomes (each unit has two potential outcomes but only one can be observed and the other is 0)

$$Y|T = 1 = (Y_1|T = 1) + \cancel{(Y_0|T = 1)} = Y_1|T = 1$$

- Adding and subtracting $\mathbb{E}[Y_0|T = 1]$, reordering and merging expectations

$$\mathbb{E}[Y|T = 1] - \mathbb{E}[Y|T = 0] = \mathbb{E}[Y_1 - Y_0|T = 1] + \underbrace{\mathbb{E}[Y_0|T = 1] - \mathbb{E}[Y_0|T = 0]}_{\text{Bias}}$$

- Association = causation + a bias term

The Bias Equation: Remarks

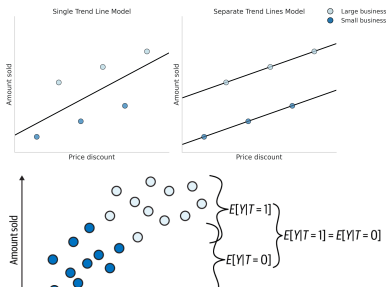
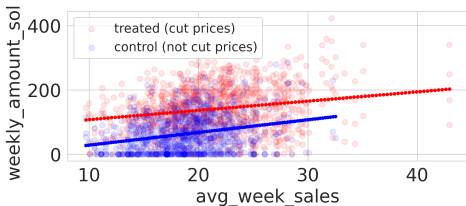
- Association = causation + a bias term

$$\mathbb{E}[Y|T=1] - \mathbb{E}[Y|T=0] = \mathbb{E}[Y_1 - Y_0|T=1] + \underbrace{\mathbb{E}[Y_0|T=1] - \mathbb{E}[Y_0|T=0]}_{\text{Bias}}$$

- Bias arises from differences between treated and control groups (besides the treatment)
 - Unobservable factors change together with treatment
 - E.g., there are confounders
 - E.g., treated and untreated businesses differ beyond price cuts
 - E.g., differ in size, location, day of the week on which they have a sale, cities they are located in
- For association to be causation, treated and control units must be **exchangeable**

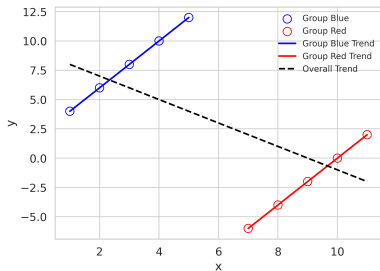
The Bias Equation: Visual Example

- Plot `weekly_amount_sold` by business size (`avg_week_sales`)
 - Color points: treatment (red) / control (blue)
 - Treated units concentrated on the right
- Difference in amount sold between groups has two causes
 - *Treatment effect*: Price cut increases amount sold
 - *Business size*: Bigger businesses sell more, can do more price cuts
- The challenge for causal inference is to untangle the two effects



Simpson's Paradox

- Trend appears in different data groups but reverses when combined
 - Relationship within subgroups can disappear or flip when aggregated
- Paradox occurs due to a confounder influencing the relationship
 - Aggregated data distorts results due to hidden variable



Simpson's Paradox: Admission bias in University

- A university has
 - Two departments: Engineering and English
 - Male overall acceptance rate: $\frac{81}{120} = 67.5\%$
 - Female overall acceptance rate: $\frac{58}{120} = 48.3\%$
- Is a bias against females, due to lower acceptance rate?
- Break down data by Engineering and English

Department	Male			Female		
	Applicants	Accepted	Rate	Applicants	Accepted	Rate
University	120	81	67.5%	120	58	48.3%
Engineering	100	80	80%	20	18	90%
English	20	1	5%	100	40	40%

- At the department level:
 - Women have higher acceptance in Engineering (90% vs 80%)
 - Women have higher acceptance in English (40% vs 5%)
- Paradox reason:
 - Engineering has high acceptance; English has low acceptance
 - More men apply to Engineering; more women apply to English
 - Women apply more to harder department, making their overall

Identifying the Treatment Effect

- We can't observe $\mathbb{E}[Y_{1i} - Y_{0i}]$ since we can only observe one of the potential outcomes Y_{1i} or Y_{0i} , but *not both*
 - How can we recover causal quantities from observable data?
- The bias equation says that:

$$\mathbb{E}[Y|T = 1] - \mathbb{E}[Y|T = 0] = \mathbb{E}[(Y_1 - Y_0)|T = 1] + \underbrace{\mathbb{E}[Y_0|T = 1] - \mathbb{E}[Y_0|T = 0]}_{\text{Bias}}$$

- If $\mathbb{E}[Y_0|T = 0] = \mathbb{E}[Y_0|T = 1]$, i.e., the treatment and the control group are comparable (exchangeable) regardless of the treatment, then:
 - The bias term vanishes $\text{bias} = 0$
 - Association is causation
 - The measured difference in means equals $ATE = \mathbb{E}[Y_1 - Y_0]$
 - The measured difference in means equals $ATT = \mathbb{E}[(Y_1 - Y_0)|T = 1]$
- **Sales Example**
 - If businesses that do and don't cut prices are similar to each other (i.e., interchangeable), then
 - The difference in amount sold between the ones that cut prices and the ones that don't, can be entirely attributed to the price cut

Exchangeability Property

- The exchangeability property is equivalent to having *potential outcomes* being independent of the treatment:

$$(Y_0, Y_1) \perp T$$

i.e., $\mathbb{E}[Y_0|T] = \mathbb{E}[Y_0]$: the treatment gives no information about the potential outcomes (i.e., $\mathbb{E}[Y_0|T=1] = \mathbb{E}[Y_0|T=0]$)

- **Note**
 - We are not asking for *observed outcome* Y and treatment T to be independent
 - This means that treatment has no effect
 - We are asking for the potential outcome to be independent of the treatment
 - The outcome observed if the subjects received the treatment is independent if it received the treatment or not
 - “Knowing whether someone received the treatment tells you nothing about what their outcome would have been without it”
 - The treatment is the only thing that causes difference in outcome between treatment and control

Phases of Causal Inference

- A causal inference problem is often broken down into two steps:
 - **Identification**
 - Express the causal quantity of interest in terms of observable data
 - Remove the bias, making treated and untreated comparable, so that all the difference can be attributed to the treatment effect

$$\mathbb{E}[Y_0|T = 0] = \mathbb{E}[Y_0|T = 1]$$

- You need to know how the treatment was distributed or assigned
 - This is why data alone (observational) cannot answer causal questions
- **Estimation**
 - Use data to estimate the causal quantity of interest:

$$te = \mathbb{E}[Y|T = 1] - \mathbb{E}[Y|T = 0]$$

Randomization Removes Bias

- Randomization of treatment forces independence to hold $\implies$ removes bias
- Assume you can randomize the treatment (e.g., you can force a business to cut prices)

$$\begin{cases} IsOnSales \leftarrow rand(t) \\ AmountSold \leftarrow f_y(IsOnSales, u_y) \end{cases}$$

- The dependency from u_t disappears since the assignment mechanism is:
 - Tied to a coin flip (unrelated to other factors in causal mechanism)
 - Fully known
- Treatment becomes independent from potential outcomes

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A/B Testing Example

- A common strategy among companies is to
 - have a cheap / free product that serves to attract customers
 - Cross-sell other products that are more profitable
 - Use email campaigns to sell premium product
 - When a customer get an email and buys the premium product, they are “converted”
- Non-randomized data, you see that:

$$\mathbb{E}[\text{Conversion} | \text{Email} = 1] > \mathbb{E}[\text{Conversion} | \text{Email} = 0]$$

- Problem: marketing team tends to send emails for customers who they think they will convert (e.g., using a survey, customers who visited the website the most)

$$\mathbb{E}[\text{Conversion}_0 | \text{Email} = 1] > \mathbb{E}[\text{Conversion}_0 | \text{Email} = 0]$$

- I.e., customers would have converted even if the email was not sent
- You need to make the treated and untreated comparable
 - E.g., you can randomize who gets the email
- Imagine that you have 3 emails

Randomized Controlled Trials

- RCTs are the most reliable way to measure causal effects
- The problem is
 - Very expensive (in time and money)
 - Potentially unethical
 - Sometimes impossible
- You can use an RCT as a benchmark
 - What is the best experiment to answer your question (even if it's not) feasible?
 - Can help discover the causal effect

–

- Standard error of the mean is related to the sample size n

$$SE = \frac{\sigma}{n}$$

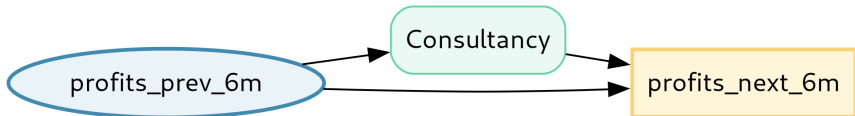
- Imagine you have data about schools in terms of number of students and average SAT score

– An frequent effect is that the top performing schools have a small

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-
- Estimate the impact of emails on conversion
 - Cross-sell email = treatment T
 - Customer converted to a new product (or not) = outcome Y

Are Consultants Worth It?

- You are a manager and you need to decide whether to hire expensive consultants
- Which one is true?
 - Do expensive consultants improve business?
 - Only very profitable businesses can afford consultants?
- Setting up a randomized trial is difficult
 - You need to untangle causation from association



Chains

- In case of $T \rightarrow M \rightarrow Y$:
 - M is “mediator” since it mediates the relationship between T and Y
 - Causation flows only in the direction of the arrows
 - Association flows both ways
- E.g., $Study \rightarrow SolveProblems \rightarrow JobPromotion$
 - Job promotion is causally dependent on studying
 - The association goes both ways
 - The more you study, the more likely to get a promotion
 - The greater the chances of promotion, the greater your chance of having studied (otherwise it would be difficult to get a promotion)
 - So $Study$ and $JobPromotion$ are
 - Dependent or
 - Not independent $Study \not\perp JobPromotion$
- If you hold the intermediary variable fixed (i.e., conditioning on M , looking for only people with the same skills at problem solving), the dependency is blocked
 - T and Y are independent given M , i.e., $(T \perp Y) | M$
 - $\mathbb{E}[Y | T, M] = \mathbb{E}[Y | M]$ and also the inverse $\mathbb{E}[T | Y, M] = \mathbb{E}[T | M]$

Forks

- In a fork $T \leftarrow X \rightarrow Y$:
 - X is a “confounder” (common cause of T and Y)
 - Association flows backward through the arrows
 - T and Y are dependent even though neither causes the other:
 $T \not\perp Y$
- E.g., $CausalKnowledge \leftarrow Statistics \rightarrow MachineLearning$
 - Assume that causal inference doesn't help with machine learning, and vice versa, so there is no edge between the 2
 - Knowing causal inference signals statistics knowledge, which predicts ML skill
- E.g., $AnswerQuestion \leftarrow GoodProgrammer \rightarrow GoodEmployee$
 - For this reason in tech interviews you are asked to solve problems that might you never find in the job you apply for (e.g., “invert a binary tree”)
- Conditioning on the common cause blocks the association:

$$T \perp Y, (T \perp Y) | X$$

- E.g., if you know that you are a good programmer (e.g., you have an impressive resume), asking you a tech question won't give new information about if you are a good employee or not

Collider

- In a collider $T \rightarrow X \leftarrow Y$:
 - X is a “collider” (two arrows collide at it)
 - T and Y are independent: $T \perp Y$
 - Conditioning on X opens the path, making T and Y dependent:
 $T \not\perp Y | X$
- E.g., $Statistics \rightarrow Promotion \leftarrow FlatteringBoss$
 - There are two ways to get a job promotion
 - You can be good at statistics
 - You can flatter your boss
 - If I don't know if you get or not the promotion (i.e., I don't condition on *Promotion*) your level of statistics and flattering are independent
 - If I know you got promoted, and I know you are bad at statistics, then you must be good at flattering your boss
 - “Explaining away” since one cause explains the effect, making the other cause less likely
- Conditioning on any *descendant* of the collider opens the path, making the variables dependent
 - E.g., $Statistics \rightarrow Promotion \rightarrow Salary \leftarrow FlatteringBoss$:
conditioning on *Salary* opens the $Statistics \perp FlatteringBoss$ path

D-separation

- A path is **blocked** given a conditioning set Z if and only if either condition holds:
 1. It contains a non-collider that has been conditioned on
 2. It contains a collider that has not been conditioned on and has no descendants that have been conditioned on
- Otherwise the path is **open** (nodes can be dependent)
- Two nodes A and B are d-separated (independent) if and only if all the paths between them are blocked

Identification

- It is hard to find the causal effect because treated and untreated are not comparable to each other, which results in bias
- Using causal graphs
 - You can understand where the bias is coming from and thus how to neutralize it
 - Thus the identification phase becomes equivalent to “close all undesirable flows of association”
- E.g., in the consultancy graph
 - There is:
 - A fork on `profits_next_6m`
 - A confounder `profits_prev_6m`
 - A causal path `consultancy` → `profits_next_6m`
 - A non-causal path (aka “backdoor path”) `profits_prev_6m` → `profits_next_6m`
 - To close the non-causal flows of association you need to condition on the confounder, i.e., `profits_prev_6m`
 - Look at a group of companies with similar past performance
 - Compare the ones that hired consultants (treated) vs the ones

Backdoor Adjustment

- You are making the “conditional independence assumption”, i.e., $(Y_0, Y_1) \perp T | X$
- I.e., units with the same level of covariates X have potential outcomes that are the same on average (it's as if they were randomized)
- Closing backdoor paths by conditioning on confounders X makes the causal quantity identifiable from observable data (aka “backdoor adjustment”)

$$ATE = \mathbb{E}_X[\mathbb{E}[Y|T=1] - \mathbb{E}[Y|T=0]] = \sum_x (\mathbb{E}[Y|T=1, X=x] - \mathbb{E}[Y|T=0, X=x])$$

- ATE is the weighted average of in-group treated-vs-control differences, weighted by group size $\Pr(X = x)$

Confounding Bias

- Confounding happens when there is an open backdoor path through which association flows non-causally, because treatment and outcome share a common cause
- Example
 - You work in HR and want to know if your new management training program is increasing employers' engagement
 - Training is optional
 - Only managers who are already doing a great job attend the program
 - The managers that need the training the most, don't attend
 - When you measure engagement of the teams, engagement under managers who took the training is much higher
 - There is a common cause between treatment and outcome, so they move together regardless of a causal effect
- You need to close all backdoor paths between treatment T and outcome Y , i.e., adjust for the common causes / adjust for confounding bias
 - E.g., control for manager's quality before taking the training
- Unfortunately you can't adjust for all common causes

Surrogate Confounding

- The confounder variable might be not be observable / measurable, but you might have measured variable that can be proxies for confounder
 - E.g., you can't measure confounder “manager quality”
 - You can measure
 - Causes
 - Manager's tenure
 - Manager's education level
 - Effects
 - Team's attrition
 - Team's past performance
- In this way you don't eliminate bias, but you can reduce it
- Partial remedies when confounders are unobserved:
 - Sensitivity analysis: Instead of asking “did I control for everything?”, ask “how strong must the unmeasured confounder be to change my conclusion?”
 - Partial identification: Use observational data to place bounds on the causal quantity rather than point-identify it
 - Randomization: Intervening on the treatment removes confounding

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- Problem in banking and lending industry: understand the impact of loan amount / credit card limits on default rate

Regression in A/B tests

- You are building a recommender system
- Goal: increase the watch time of the streaming service
- A/B test
 - 1/3 but representative fraction of your customer base gets the new system
 - The rest gets the old version
 - Collect data
 - Because of the randomization you can assign a causal meaning to the use of treatment
 - Compute the average watch time per day + std err
- You can compute all at once with a regression

$$WatchTime_i = \beta_0 + \beta_1 I(new)_i + \varepsilon_i$$

where

- $\hat{\beta}_1$ is the increase in watching time, i.e., ATE
- This is exactly equivalent to mean and std dev of *WatchTime*

Adjusting with Regression

- Effect of credit lines on default risk
- Data has:
 - Binary variable representing if a customer has defaulted or not
 - Customers' features to indicate credit worthiness (e.g., monthly wage, credit scores, tenure current company)
- Model the problem as regression

$$Default_i = \beta_0 + \beta_1 limit_i + \varepsilon_i$$

where

- $\hat{\beta}_1$ is the amount of risk increase given 1 USD increase in limit
- Computing the regression you see that the coefficient is negative
 - There is confounding (since banks give higher rate to trustworthy customer)
 - Doesn't have a causal interpretation
- Instead of adjusting manually for each confounder, you can add the confounders to the regression

$$Default_i = \beta_0 + \beta_1 limit_i + \sum_j \theta_j X_{ji} + \varepsilon_i$$

- E.g., X params can be wage, credit scores, etc

Regression theory

- The vector of β^* minimizes the mean squared error

$$\beta^* = \operatorname{argmin}_{\beta} \mathbb{E}[(Y_i - X_i^T \beta)^2]$$

which can be estimated as $\hat{\beta} = (X^T X)^{-1} X^T Y$

- In case of a single variable, regression estimates how treatment and outcome move together (covariance), scaling this by the variance of units of treatment
- In case of multivariate regression, each coefficient is the coefficient of the regressor (bivariate regression), after accounting for the effect of other variables in the model
 - You can regress against all the confounders and extract the residual
- You can extrapolate the model outside the region where you have data
 - It is risky since you never know what happens in region where you don't have data points

FWL Theorem

- Make non-experimental data look as if the treatment was randomized (de-biasing)
- Estimating a multi-variate regression is equivalent to the 3 steps:
 1. Denoising: Regress outcome Y on the confounder variables to obtain the outcome residuals $\tilde{Y}$
 2. Debiasing: Regress the treatment T on confounders X to obtain the treatment residuals $\tilde{T}$
 - Version of the treatment that is uncorrelated with the variables used in debiasing the model
 3. Regress outcome residuals $\tilde{Y}$ on treatment residuals $\tilde{T}$ to obtain estimate of causal effect of T on Y

Non-linear

- You want to estimate the causal effect of credit limit on credit card spend, using wage as only confounder
 - It can be that the relationship is not linear
 - E.g., the effect of credit lines on spend diminishes as you increase credit lines
- You can apply some transform to X (e.g., square root, log)
- Then you can apply the usual flow (adding confounders, FWL theorem)

Regression

- You need to assume that treatment looks randomly assigned once certain covariates are accounted for
 - You can't know if all confounders have been included in the model
 - Only a randomized controlled experiment can confirm it
 - E.g., giving random credit lines (including very high) to risky customers who will likely default and cause huge loss
- The best next thing to a randomized controlled trial is a “stratified random experiments”
 - Create multiple local experiments, drawing from different distributions, depending on covariates
 - E.g., instead of randomizing credit lines completely, you use features (e.g., credit score, which is a proxy for customer risk) to create groups of customers (e.g., high-risk, neutral, low-risk)
 - Then for the high-risk, you randomize credit lines from a distribution with low average
 - For the low-risk, you randomize credit lines from a distribution with higher average
- Stratified random experiments are:
 - cheaper
 - more complex than completely random experiments

Dummy Variables

- Problem: Simple regression of outcome on treatment yields biased estimates with confounders
 - E.g., estimating `default` `credit_limit` gives negative coefficient (counterintuitive)
 - Occurs due to imbalanced treatment assignment across confounding groups—lower-risk customers got higher credit lines
- Solution: Dummy Variables (One-Hot Encoding):
 - Create binary columns for each categorical group (1 if in group, 0 otherwise)
 - Drop first dummy to avoid multicollinearity
 - Include dummy controls in model: `default` `credit_limit` + `C(credit_score1_buckets)`
- Result after adding dummies:
 - Model estimates one intercept per group, shared slope across all groups
 - Shared slope is variance-weighted average of regression slopes within each group

Saturated Regression Model

- Definition: A saturated regression model includes interaction terms between treatment and dummy variables to estimate separate effects for each group
 - Created by interacting treatment with categorical variables:
`treatment * C(categorical_var)`
 - Each group gets its own intercept and slope
- Running a saturated model produces identical results to:
 - Segmenting data by groups manually
 - Estimating treatment effect separately in each group
 - Computing a weighted average (weights = group size)
 - E.g., both approaches yield the same Average Treatment Effect (ATE)
- Model interpretation:
 - Treatment coefficient represents effect in the omitted (reference) group
 - Interaction parameters show how effects differ from the reference group
 - Sum interaction term + treatment coefficient to get effect for that specific group
- Trade-off: More parameters $\rightarrow$ more variance

Omitted Variable Bias and Regression

- The bias due to omitting a variable is equal to tA

Neutral Controls: Controlling Variance Without Changing Bias

- **Problem:** Should we include every available variable in our regression model?
 - Some variables are confounders (must include to eliminate bias)
 - Some variables don't cause either treatment or outcome
 - Question: Which additional variables help or hurt precision?

Noise-Reducing Controls

Variables that *predict the outcome well* but aren't confounders

- *Example:* Credit score bucket $\rightarrow$ credit limit $\rightarrow$ default
 - `credit_score1_buckets`: confounder (must include)
 - `credit_score2`: predicts default but not treatment
- *Effect:* Reduces variance in ε_i by FWL denoising step
- *Result:* Same $\hat{\beta}_{\text{treatment}}$, much tighter confidence intervals

Noise-Inducing Controls

Variables that *predict the treatment well* but not the outcome

- *Example:* Conditionally random spend experiment
 - `credit_score1`: drives treatment assignment but $\perp$ outcome
 - Incorrectly including it increases variance
- *Effect:* Reduces treatment variance by explaining away $\text{Var}(T)$

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 - *Feature Selection Dilemma*
- Propensity score
- Effect heterogeneity
- Metalerners
- Difference-in-differences
- Synthetic control
- Geo and switchback experiments
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- **Reality**: Most variables are imperfect confounders
 - Strong cause of T , weak cause of Y (like noise-inducing)
 - Strong cause of Y , weak cause of T (like noise-reducing)

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The Impact of Management Training

- A common phenomenon in tech companies is talented individual contributors (ICs) transitioning to management tracks
 - This transition often requires very different skill sets and comes at high personal cost for the new managers and their teams
 - Multinational tech company invest in manager training programs for new managers to ease this transition

The problem of non-compliance

- The company
 - Selects managers randomly for the training program to measure its effectiveness
 - Compares employee engagement scores between teams whose managers received training vs those who didn't
 - Estimates an unbiased average treatment effect (ATE)
- **Non-compliance**, i.e., people not receiving the treatment as assigned
 - Some assigned managers didn't show up for training
 - Other managers obtained the training without being assigned to it
- **Result**: the randomized study became observational in nature
- This introduced bias into the causal estimates, requiring adjustment for confounders

Naive Comparison

- Comparing treated vs control without adjustment shows $ATE \approx 0.44$
 - This is likely biased upward because managers with already-engaged teams are more likely to self-select into training
- Controlling for all available confounders via regression reduces the estimate to $ATE \approx 0.27$
 - **Positive confounding bias:** managers whose employees were already more engaged were more likely to have participated in the training program

Propensity score

- **Propensity score**: Conditional probability of receiving treatment given covariates

$$e(x) = \Pr(T|X)$$

- Acts as dimensionality reduction
 - Instead of conditioning on high-dimensional X , condition on single propensity score
- Blocks backdoor paths through confounders and balance

$$(Y_1, Y_0) \perp T | \Pr(x)$$

- **Intuition**
 - If two managers have same propensity score but different treatment status, difference is due to chance
 - Treatment becomes “as good as random” after controlling for propensity score

Propensity Score Estimation

- Can use ML models to estimate propensity score
 - Out-of-fold predictions to avoid overfitting bias
- Binary treatment makes
 - Use logistic regression to estimate $e(x)$ from covariates
 - Replace true (unknown) propensity score with estimated version
 - Apply in observational data where assignment mechanism unknown

Propensity Score and Linear Regression

- Freilich-Wooldridge-Newey (FWL) theorem: linear regression models

$$\mathbb{E}[T|X]$$

in debiasing step

- Propensity score estimation achieves similar orthogonalization
- Can use estimated propensity score $\hat{e}(X)$ in regression:
 - E.g., `engagement_score` `intervention` + `propensity_score`
- Both approaches orthogonalize treatment but use different models for T

Propensity score matching

- Matches treated and control units based on propensity score similarity
- Essentially 1-Nearest Neighbors (i.e., KNN with $K=1$) on propensity score
 - Fit KNN on treated units to impute Y_1 for controls
 - Fit KNN on control units to impute Y_0 for treated
- Matched value is outcome of nearest unit

$$A\hat{T}E = \frac{1}{N} \sum ((Y_i - Y_{jm}(i))T_i + (Y_{jm}(i) - Y_i)(1 - T_i))$$

- Cons
 - KNN inefficient in high dimensions
 - Difficult to derive variance

Inverse Propensity Weighting (IPW)

- **Core Principle**
 - Reweight data by inverse probability of observed treatment
 - Create pseudo-population where treatment appears randomly assigned
 - For treated: upweight those unlikely to be treated (low $e(x)$)
 - For control: upweight those likely to be treated (high $e(x)$)

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